Experimental Evaluation of CAN, UART, and I2C Communication in a Baja SAE Prototype

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***Abstract*—Distributed data-acquisition systems in Baja SAE prototypes, like distributed sensing and actuation in mobile robotic platforms, depend on communication links that periodically exchange information through harnesses exposed to electromagnetic disturbances. This work experimentally evaluates I2C, UART, and Classical CAN/TWAI on ESP32 microcontrollers using a common request-response protocol with integrity checks at 100 Hz. The tests considered link length, engine operation, and intensified coupling to the ignition system. I2C operated adequately over the short link but failed at the meter-scale link. UART and CAN operated correctly under normal routing; under the severe condition, UART exhibited losses and update discontinuities, whereas CAN preserved application-level message integrity. Under the tested conditions, the evaluated CAN implementation was the most suitable main-bus candidate for the proposed architecture, while I2C and direct UART remain appropriate as local interfaces within compatible electrical and topological constraints.**

***Index Terms*—Baja SAE, CAN bus, data acquisition, distributed embedded systems, electromagnetic interference, I2C, mobile robotics, UART.**

I. Introduction

The Baja SAE competition, associated with the Society of Automotive Engineers (SAE), challenges university teams to design, manufacture, and validate a single-seat vehicle intended for off-road operation [1]. In this environment, onboard instrumentation makes it possible to relate mechanical changes to measurements of rotational speed, vehicle speed, temperature, acceleration, position, and electrical quantities. Data-acquisition and telemetry systems have already been developed specifically for Baja SAE vehicles, demonstrating the relevance of these resources for experimental validation [2]–[4]. As the number of sensors increases, a strictly centralized architecture increases harness length, connector count, and fault-localization difficulty. Distributing acquisition among electronic units located near the sensors reduces point-to-point wiring, but makes the interface between nodes a design variable. In-vehicle networks are used precisely to meet reliability, scalability, and timing requirements in distributed electronic systems [5], [6].

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Fig. 1. UFV Baja Pererecas prototype used as the application context for the onboard communication network. Photograph: UFV Baja Pererecas team archive.

The UFV Baja Pererecas prototype used as the application context is shown in Fig. 1.

Reliable inter-node communication supports distributed sensing, control, and actuation in mobile robotic platforms. A suitable communication layer can reduce point-to-point wiring and support modular expansion of the robotic platform [7], [8]. This study does not evaluate autonomy or closed-loop performance; its contribution is the experimental validation of communication infrastructure that can support such higher-level functions.

The objective of this work is to experimentally evaluate the Inter-Integrated Circuit (I^2C), direct 3.3 V Universal Asynchronous Receiver/Transmitter (UART), and Controller Area Network (CAN) using the ESP32 Two-Wire Automotive Interface (TWAI) under Baja SAE conditions, with a common request-response protocol, sequence control, integrity verification, and 100 Hz traffic. Independent logs distinguish initialization unavailability, transactional failures, and update discontinuities; the comparison therefore concerns complete communication implementations. The main contributions are a

controlled common-protocol comparison, a repeated-run evaluation under intensified ignition coupling, and an implementation-oriented assessment for a distributed mobile platform.

II. Related Work

Previous Baja SAE studies have developed data-acquisition and telemetry systems, including CAN-based architectures [2]–[4]. Hubbard and Mynderse validated a vehicle-level acquisition system on the bench and in the vehicle, but did not compare communication interfaces [2]. Broader reviews discuss reliability, timing, and architectural tradeoffs in in-vehicle networks [5], [6].

In robotics, distributed architectures likewise rely on communication backbones connecting sensors, actuators, and controllers. Bonarini *et al.* discuss this modular view [7], and Losada *et al.* demonstrate a CAN-based architecture for mobile robots [8]. The literature reviewed here does not provide the same controlled hardware comparison of I^2C, direct 3.3 V UART, and CAN using a common application-layer protocol, repeated runs, and ignition-related electromagnetic stress. This work addresses that gap at the communication-infrastructure level rather than as a demonstration of autonomy.

III. Experimental System and Methodology

A. Communication interfaces and technical characteristics

The three evaluated communication interfaces are based on distinct electrical and architectural principles. Their suitability for a distributed network depends not only on the transmission rate, but also on the supported topology, medium-access mechanism, error-detection capabilities, and resources required to add new modules.

I^2C: the *Inter-Integrated Circuit* is a synchronous, addressed bus in which the serial-data (SDA) and serial-clock (SCL) lines can be shared by controllers and devices. Its outputs are *open-drain*: circuits actively force the low level, while *pull-up* resistors establish the high level. After each byte, the receiver indicates the outcome of the operation through an *Acknowledge* (ACK) or *Not Acknowledge* (NACK). Although multiple devices can share the same bus, I^2C has no universal maximum length. Its practical reach depends mainly on total capacitance, rise times, cable characteristics, stubs, frequency, and pull-up resistance, and the specification is typically associated with capacitances of up to 400 pF. Therefore, it is better suited to short interconnections within an electronic unit than to meter-scale vehicle harnesses [9].

UART: the *Universal Asynchronous Receiver/Transmitter* provides asynchronous serial communication using start and stop bits for synchronization. In this work, a direct 3.3 V connection with crossed TX and RX lines formed a point-to-point *full-duplex* link. The allowable length depends on the associated electrical layer, voltage level, bit rate, capacitance, and noise environment. UART also provides no native addressing, arbitration, retransmission, or concurrent-access control. Its use in *multidrop* networks requires external transceivers, such as RS-485, as well as a higher-layer protocol, direction control, and mechanisms to coordinate simultaneous transmissions. Therefore, distribution is not a native feature of the evaluated direct UART implementation, but depends on additional hardware and protocol resources [10], [11].

CAN/TWAI: the *Controller Area Network* is a differential, multi-master, message-oriented bus. Frame identifiers also establish priority during nondestructive arbitration, allowing several nodes to share the medium without destroying the highest-priority message. The protocol incorporates a cyclic redundancy check (CRC), acknowledgment, active error signaling, automatic retransmission, and fault confinement. On the ESP32, the controller compatible with Classical CAN is named the *Two-Wire Automotive Interface* (TWAI). Because the ESP32 integrates the CAN/TWAI controller but not the external bus transceiver, an external component is required to convert the controller TX/RX signals into the differential CANH/CANL levels [12]–[14]. This work used the 3.3 V SN65HVD230, which is compatible with the ESP32 and supports signaling rates of up to 1 Mbit/s, as well as thermal protection and a high-impedance state when unpowered [15]. CAN reach depends on the bit rate, propagation delays, topology, stubs, and termination. Unlike point-to-point UART, additional modules can share the same CANH/CANL pair through message identifiers, provided the electrical limits of the network are respected.

B. Target application and experimental architecture

The proposed distributed architecture, shown in Fig. 2, consists of two acquisition units and a dashboard interconnected by CAN. One *data logger* concentrates signals near the engine, battery, and fuel tank, while the second receives vehicle-speed, continuously variable transmission (CVT) temperature, inertial measurement unit (IMU), and Global Positioning System (GPS) data and transmits telemetry to the pit station through E32 LoRa modules. The dashboard receives critical variables. This organization motivated the comparison of communication interfaces before implementation of the complete network.

In the test bench, two ESP32 DevKit V4 boards were configured as transmitter and receiver, with only one interface active in each run. For I^2C, SDA, SCL, and a common ground were used with 2.2 kΩ pull-up resistors selected to reduce rise times in the presence of the additional harness capacitance [9]. UART used crossed TX and RX lines, a common reference, and 33 Ω series resistors placed near the transmit outputs to damp oscillations and reflections. For CAN, each ESP32 was connected to an SN65HVD230 transceiver; CANH and CANL were twisted and terminated with 120 Ω at both ends [15].

The experiments compare complete communication implementations rather than isolated protocol algorithms. In particular, the evaluated UART link uses direct 3.3 V single-ended signaling at 115.2 kbit/s, whereas CAN uses SN65HVD230 differential transceivers at 500 kbit/s. Consequently, differences observed between UART and CAN may reflect both protocol-level mechanisms and physical-layer characteristics; the tests were intended to support an implementation choice for the vehicle, not to isolate those effects experimentally.

Fig. 3 compares the routing geometries with the physical setup, while Table I presents the main parameters.

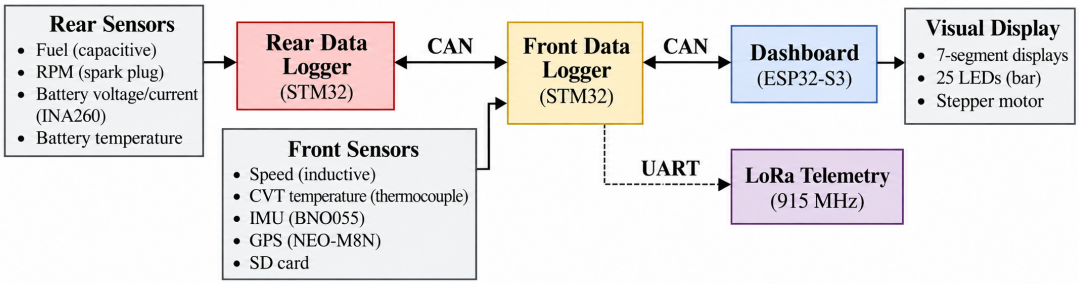
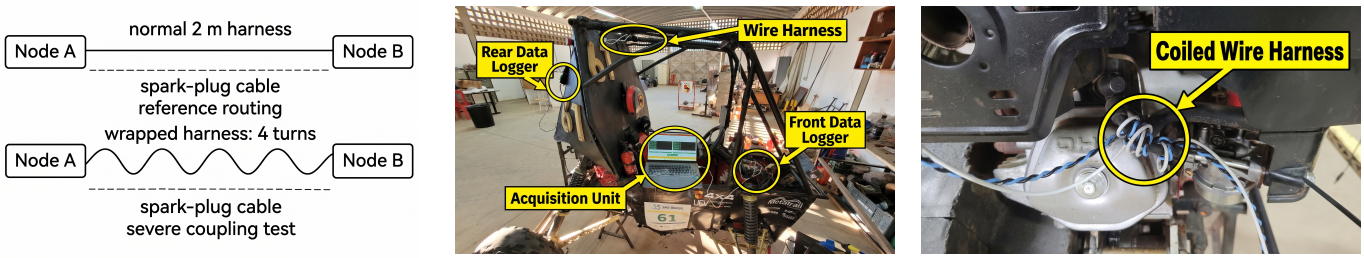


Fig. 2. Proposed distributed electronic architecture. Sensors are grouped into acquisition units located near their installation regions. The acquisition units and the dashboard share the CAN bus, while telemetry uses an independent LoRa link. Diagram: prepared by the authors.



(a) Evaluated routing geometries. (b) Actual experimental setup. (c) Severe condition: four turns.

Fig. 3. Comparison between the evaluated routing geometries and the actual experimental setup. Panel (a) presents the reference routing and the severe condition planned for the harness. Panel (b) shows the assembled test bench, and panel (c) presents the severe condition, in which the 2 m harness was wrapped four times around the spark-plug cable while the same geometry was maintained across repetitions. This configuration was used only as a stress test and does not represent the recommended routing in the vehicle. Diagrams: prepared by the authors. Photographs: authors' archive.

TABLE I
MAIN TEST CONFIGURATIONS.

Interface	Physical rate	Length	Signal conditioning
I^2C	100 kHz	16.8 cm and 2 m	2.2 kΩ
UART	115,200 bit/s	2 m	33 Ω series
CAN/TWAI	500 kbit/s	2 m	120 Ω at both ends

The evaluated lengths were selected according to the intended role of each interface. The 16.8 cm I^2C link represents a local interconnection within or near an electronic unit and also serves as an operational control, whereas the 2 m link represents communication between spatially distributed vehicle nodes. UART and CAN were considered primarily as inter-node candidates and were therefore evaluated at 2 m. After I^2C failed to initialize at 2 m, the short link was repeated as a post-control; because the meter-scale link was already unavailable before the ignition-stress phase, I^2C was not subjected to the engine-on or wrapped-harness tests. Thus, the experiment does not constitute a symmetric distance-response comparison among all three interfaces.

Under reference routing, the harness was kept away from the spark-plug cable; under the severe condition, the same 2 m harness was wrapped four times around it, creating a fixed and reproducible intensified-coupling geometry that does not represent a typical installation or a normative recommendation. The campaigns were executed sequentially rather than randomized or interleaved: I^2C was evaluated first, followed by UART and CAN under normal routing, with engine-off runs preceding engine-on runs; the wrapped-harness stress campaigns were then performed, first for UART and then for CAN. Tests with the engine off served as controls for the routing geometry, whereas engine-on tests evaluated the combination of that geometry with ignition activity. The UART bench condition was retained as an additional reference.

C. Application protocol and procedure

The request frame contains eight bytes: message type, a 16-bit sequence number, sequence-derived patterns, a fixed marker, and a Modbus CRC-16. The response confirms the sequence and reports the receiver status. START, DATA, and END messages control initialization, the data phase, and termination. Logs were stored in RAM during each test and exported after transmission, reducing interference from serial output on timing.

Each formal test consisted of 10,000 transactions, one every $T_s = 10$ ms. The receiver was started before the transmitter. After START confirmation, the DATA phase was executed; when synchronization was not obtained within the defined window, the data phase was not started. The transmitter and receiver generated independent files that were subsequently correlated by sequence number.

D. Communication metrics and statistical analysis

The packet error rate (PER) and packet delivery ratio (PDR) were calculated, respectively, as

$$\mathrm{PER} = \frac{N_{\mathrm{failures}}}{N_{\mathrm{attempts}}}, \qquad \mathrm{PDR} = \frac{N_{\mathrm{correct}}}{N_{\mathrm{attempts}}} = 1 - \mathrm{PER}, \quad (1)$$

where N_{failures}, N_{attempts}, and N_{correct} denote failed, attempted, and correctly completed transactions, respectively.

For transaction i, the application-level round-trip time was defined as

$$\mathrm{RTT}_i = t_{\mathrm{ACK},i} - t_{\mathrm{REQ},i}, \quad (2)$$

where $t_{\mathrm{REQ},i}$ is the request start time and $t_{\mathrm{ACK},i}$ is the time at which the corresponding response was validated by the transmitter.

For updates recognized by the receiver, jitter relative to the nominal period $T_s = 10$ ms was calculated as

$$J_i = \left| \Delta t_i - T_s \right|, \quad (3)$$

where Δt_i is the interval between consecutive updates. Update discontinuity and transaction-deadline violation were represented by the indicator functions

$$D_i = \mathbb{I}(\Delta t_i > 1.5 T_s), \qquad M_i = \mathbb{I}(\mathrm{RTT}_i > T_s). \quad (4)$$

Timeouts do not produce valid RTT samples and are therefore treated separately from M_i; failed transactions may nevertheless yield $D_i = 1$ when the next update arrives approximately two periods later.

The experimental unit was each complete run rather than each packet. Each primary UART/CAN condition comprised ten independent runs, with 10,000 transactions per run; therefore, the packet count characterizes within-run behavior but does not increase the independent sample size beyond $n = 10$. Approximately symmetric run-level metrics were summarized by mean and standard deviation, whereas maximum jitter was summarized by the median, interquartile interval $[Q_1, Q_3]$, and range. Packet-error proportions were accompanied by 95% Wilson intervals [16], and zero-error conditions were additionally interpreted using the approximate rule-of-three upper bound [17].

Confirmatory inference was performed at the run level. H1 compared the proportion of affected spark-plug-wrapped UART runs with the engine on versus off, and H2 compared UART and CAN under the wrapped, engine-on condition. Two-sided Fisher exact tests were used for the proportions [18], the absolute risk difference was accompanied by a 95% Newcombe interval [19], and the two confirmatory p-values were adjusted using Holm's procedure [20]. RTT and jitter comparisons were treated as descriptive or exploratory because of the limited number of independent runs and the different physical-layer rates. Normal communication failures remained valid outcomes; a run would be excluded only because of a documented external intervention.

IV. Results

A. I^2C availability and integrity

The formal I^2C results are presented in Table II. The 16.8 cm link synchronized in every run both during the initial baseline and in the subsequent control, whereas no 2 m run started the DATA phase. Recovery after returning to the short link indicates that the unavailability was associated with the evaluated physical configuration rather than with a permanent node failure.

As a complementary verification at 25 kHz, the short link completed all 30 read-and-write cycles, whereas the 2 m link completed none. Because only one run was performed for each length, this result was treated as descriptive only.

B. UART and CAN reliability

For UART, ten runs were conducted under each condition. In the bench condition, under normal routing with the engine off and on, and under the wrapped condition with the engine off, all 100,000 of 100,000 transactions were completed without failures or affected runs. With the harness wrapped and the engine on, 99,987 of 100,000 transactions were completed; the 13 failures affected eight of the ten runs and were classified at the transmitter as ten *timeouts* and three CRC errors. At the receiver, nine sequence gaps and 104 discarded bytes were observed. Because these counters may represent different manifestations of the same event, they were not summed.

For CAN, ten runs were conducted under each reported condition. Normal routing completed 100,000 of 100,000 transactions with the engine off and another 100,000 of 100,000 with the engine on. Under the wrapped conditions with the engine off and on, 100,000 of 100,000 transactions were also completed in each group. No CAN condition exhibited application-level losses, *timeouts*, sequence gaps, *bus-off* events, or affected runs. Under the severe condition, seven internal bus-error events were recorded in two runs, but they remained contained by the data-link layer [12]–[14].

C. RTT, jitter, and descriptive statistics

Table III summarizes the timing metrics at the run level. For each protocol, RTT and p99 jitter remained practically stable between normal routing and the wrapped condition. The absolute differences between UART and CAN should not be interpreted as intrinsic protocol properties because different physical-layer rates were used.

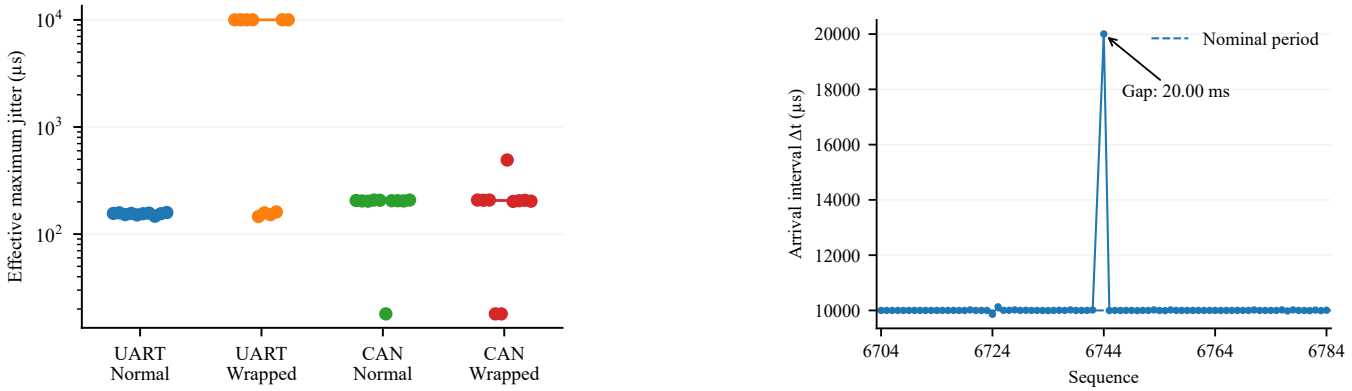
The main temporal degradation occurred during rare UART events under the severe condition: six runs exhibited nine discontinuities, with update intervals close to 20 ms and a bimodal distribution of maximum jitter. CAN exhibited no discontinuities. Fig. 4 illustrates the distribution across runs and one of the gaps observed in UART.

All valid RTTs remained below 10 ms, with maxima of 1.938 ms for UART and 0.995 ms for CAN under the wrapped, engine-on condition; transaction-deadline counters were zero. For deadline accounting, timeouts were additionally counted as missed deadlines: UART had 10 timeout-based deadline misses in 100,000 transactions, whereas CAN had none.

TABLE II

FORMAL I²C RESULTS AT 100 KHZ. RUN-LEVEL VALUES ARE MEAN ± STANDARD DEVIATION; RTT SUMMARIZES THE PER-RUN MEDIANS.

Condition	Synchronization	Correct/attempted	PER (95% CI)	Failures/run	Median RTT/run (μs)	Jitter p99 (μs)
16.8 cm, baseline	10/10	99,989/100,000	0.011% [0.0061, 0.0197]	1.10 ± 1.29	2068.4 ± 1.7	6.6 ± 1.4
2 m, formal	0/10	not applicable	not applicable	not applicable	not applicable	not applicable
16.8 cm, control	10/10	99,999/100,000	0.001% [0.0002, 0.0057]	0.10 ± 0.32	2067.5 ± 2.6	5.6 ± 1.9



(a) Effective maximum jitter per run.

(b) Example of the first UART gap.

Fig. 4. Temporal continuity with the engine on. Panel (a) shows the effective maximum jitter across independent runs, while panel (b) illustrates a UART update gap of approximately 20 ms under the severe coupling condition.

TABLE III

RUN-LEVEL TIMING STATISTICS WITH THE ENGINE ON. RTT AND P99 JITTER ARE MEAN ± STANDARD DEVIATION; MAXIMUM JITTER IS MEDIAN $[Q_1, Q_3]$. DISCONT. REPORTS AFFECTED RUNS/TOTAL EVENTS.

Interface	Installation	n	Mean RTT (μs)	RTT p99 (μs)	Jitter p99 (μs)	Maximum jitter (μs)	Discont.
UART	normal routing	10	1869.8 ± 0.3	1902.1 ± 1.0	123.4 ± 1.8	155.5 [152.8, 156.8]	0/0
UART	spark-plug wrapped	10	1869.8 ± 0.6	1901.7 ± 0.7	123.6 ± 1.0	10016.5 [158.8, 10022.5]	6/9
CAN	normal routing	10	525.0 ± 2.0	686.5 ± 16.4	12.5 ± 2.0	205 [204, 206.8]	0/0
CAN	spark-plug wrapped	10	522.8 ± 7.5	672.4 ± 41.9	12.4 ± 2.0	206 [202.2, 207.8]	0/0

TABLE IV

CONFIRMATORY TESTS AT THE RUN LEVEL.

Hypothesis	Affected runs	Risk diff. (95% CI)	p; p_{Holm}
H1: UART wrapped, ON vs. OFF	8/10 vs. 0/10	0.80 [0.38, 0.94]	0.000714; 0.001429
H2: UART vs. CAN, wrapped ON	8/10 vs. 0/10	0.80 [0.38, 0.94]	0.000714; 0.001429

D. Confirmatory tests

Table IV summarizes the confirmatory tests. Both comparisons remained significant after Holm correction. Under the tested sequence, the proportion of affected runs was higher for spark-plug-wrapped UART with the ignition active than inactive, and higher for UART than CAN under the wrapped, engine-on condition.

V. DISCUSSION

Normal routing did not differentiate the evaluated UART and CAN implementations in application-level integrity: both completed all 2 m transactions with the engine off and on. The wrapped condition with the engine off also had no failures and served as a geometry control. Degradation appeared only when intensified coupling was combined with active ignition, consistent with greater susceptibility of the evaluated direct 3.3 V single-ended UART link. The experiment does not isolate protocol- from physical-layer effects: CAN used differential SN65HVD230 transceivers at 500 kbit/s, whereas UART used direct single-ended signaling at 115.2 kbit/s. The results therefore support the complete CAN implementation for the target vehicle rather than an intrinsic protocol-only superiority claim.

Although aggregate UART PER under the severe condition was only 0.013%, failures affected 80% of independent runs, with a risk difference of 0.80 and lower confidence limit of 0.38. Nine losses also produced update intervals near 20 ms, so PER alone does not capture temporal impact. Because maximum UART jitter was bimodal, the median, interquartile range, range, and affected-run count are more informative than a mean alone. CAN recorded seven internal error events in two runs but no application-level loss; these records may reflect common underlying disturbances and should not be interpreted as seven independent physical events.

I²C exhibited a different failure mode: the 2 m link did

not reach the DATA phase, whereas both 16.8 cm controls remained operational. This supports an association with the evaluated I^2C electrical configuration rather than permanent node damage or an exclusively high-level software fault, but does not establish a universal distance limit. Because cable lengths were intentionally asymmetric, the study is not a distance-response comparison among all three interfaces: 16.8 cm represents a local I^2C use case and control, while 2 m represents the distributed inter-node scale.

Inference should be interpreted at the run level. Each main UART/CAN condition contained ten independent runs; the 10,000 transactions per run characterize within-run behavior but are not independent replicates. The single-run I^2C diagnostics at 25 kHz are descriptive, and sequential rather than randomized test blocks leave slow environmental or vehicle changes as possible confounders. No direct electromagnetic-field measurement was performed.

Computational cost was not instrumented. TWAI and UART provide hardware resources, but sequence checking, the common CRC-16, and deadline control remained in software [10], [14]; quantitative processor-usage comparisons therefore require direct instrumentation.

For the proposed architecture, CAN is additionally attractive because message-oriented identification, nondestructive arbitration, priorities, multi-node operation, and fault confinement support expansion beyond the two-node test. These properties also match communication-backbone requirements in distributed robotic systems [7], [8]. The present work validates this lower-level infrastructure, not navigation, autonomy, or closed-loop performance. Direct UART remains suitable for local point-to-point links, and I^2C for short local interconnections within their electrical and topological constraints.

VI. Conclusion

A common ESP32 platform compared I^2C, direct UART, and CAN/TWAI using eight-byte frames, CRC-16, sequence control, 100 Hz traffic, and independent logs. I^2C operated over 16.8 cm but failed to establish communication in all ten 2 m runs, with operation recovered in the short-link post-control. Under normal 2 m routing, UART and CAN completed all transactions with the engine off and on. Under intensified ignition coupling, UART exhibited 13 failures across eight of ten runs and nine update discontinuities, whereas CAN delivered all 100,000 application-level transactions despite internal error events. The run-level risk difference was 0.80 [0.38, 0.94].

For the evaluated architecture, CAN was the most suitable main-bus candidate because it combined robust application-level delivery in the severe test with a scalable multi-node bus. From a mobile-robotics perspective, these results support a communication layer connecting distributed sensing and actuation; higher-level robotic tasks were outside the scope. This conclusion applies to the complete implementations: UART and CAN differed in signaling and physical rate, while I^2C served local and meter-scale roles. I^2C and UART remain relevant local interfaces when electrical, length, topology, and electromagnetic conditions are compatible.

Limitations include two nodes, specific harness geometries, low bus load, ten independent runs per main condition, sequential test blocks, no direct electromagnetic-field measurement, and single-run I^2C diagnostics at 25 kHz. Future work will integrate the complete CAN network, increase bus load, port the implementation to STM32, instrument processor use and energy consumption, and evaluate distributed sensing and control in a mobile robotic context.

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